Intro to Machine Learning

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A field of study where we create computer programs that learns from the data.

Ex- House Price Prediction Software

Email Detection Software

Car Price Prediction Software, etc.

For every example / project, data differs

* Difference b/w AI, ML & DL :-

AI → Making machines behave intelligently

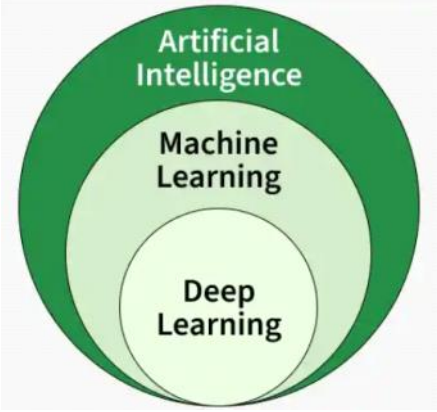
Ex- Chatbot, Self driving car, voice assistants, etc.

ML → Subset of AI where machines learn patterns from the data

Ex- Email detection, etc.

DL → Subset of ML used for performing complex tasks.

Ex- Face Recognition, etc.



* Machine Learning :-

* Machine Learning :-

↳ creating models

↓

software

↓

Computer program

↓

Algorithm ⇝ set of rules / instructions [can have mathematical eq.]

↓

In ML, we use multiple Algorithms

↓

which algorithms ??

Types of ML :-

1. Supervised ML
2. Unsupervised ML

Ex- Conduct a test of science of grade 5 students.

↙ Questions ↘ Answers

for practice

On the examination day → Question

↳ student will be able to answer because it has practiced on provided questions & answers

Similarly, providing question & answer (i/p & o/p) to the ML model is known as Supervised learning.

Ex- Car Price Prediction

Supervised ML → Type of ML when we provide both i/p & o/p to the model [Labelled data]

Unsupervised ML → Type of ML where we provide only i/p [unlabelled data]

Supervised & Unsupervised ML Algorithms :-

Supervised	Unsupervised
Linear Regression	Clustering
Logistic Regression	↳ K-Means
KNN	K- Means ++
Decision Trees	Hierarchical Clustering, etc.
Bagging / Boosting	
Naive Bayes	
SVM	

Some other unsupervised techniques :-

- Dimensionality Reduction
- Anomaly Detection